

MACHINE LEARNING ITA0609

SDG ASSIGNMENT

Name: SHAIK THOUHID

Register Number: 192421280

Course Code: ITA0609

Overfitting in Deep Neural Networks and Regularization for Rare-Disease Outbreak Prediction

SDG 3 - Good Health and Well-being

Assignment Question

Explain overfitting in deep neural networks and describe two regularization techniques suitable for a small dataset predicting rare disease outbreaks in low-resource regions.

1. Introduction

Deep neural networks can identify complex relationships in health data such as climate conditions, reported symptoms, population movement, sanitation indicators and previous outbreak records. However, when the available dataset is small, the model may memorize the training examples instead of learning general patterns. This problem is called overfitting. It is especially serious in rare-disease outbreak prediction because positive outbreak cases are limited and health records from low-resource regions may be incomplete, imbalanced or noisy.

A reliable model must perform well not only on the data used for training but also on new regions, future time periods and unseen outbreak situations. Regularization methods reduce unnecessary model complexity and improve the ability of the network to generalize.

2. Meaning of Overfitting

Overfitting occurs when a deep neural network learns the training dataset too closely, including random noise, recording errors and accidental patterns. The model then shows very high training accuracy but poor validation or testing performance.

For example, a network may learn that a rare outbreak occurred only in one particular village during the training period. Instead of learning the actual risk factors such as rainfall, mosquito density, water contamination or vaccination coverage, it may associate the village name itself with the outbreak. When applied to another village, its prediction becomes unreliable.

3. Main Causes of Overfitting in This Problem

- Small dataset: Rare diseases produce only a limited number of confirmed outbreak records.
- Too many network layers or neurons: A highly complex model can memorize small training samples.
- Class imbalance: Non-outbreak records may greatly outnumber outbreak records.
- Noisy and missing data: Low-resource regions may have incomplete reporting, delayed diagnosis and inconsistent measurements.
- Too many training epochs: Repeated training can make the model fit the training data excessively.
- Non-representative samples: Data from a few locations may not represent all regions.
- Data leakage: Future information or duplicate patient records may accidentally enter the training set.

4. Signs Used to Detect Overfitting

Observation	Training Result	Validation/Test Result
Accuracy	Very high	Much lower
Loss	Continues decreasing	Starts increasing
Predictions	Correct on known samples	Unstable on new regions
Confidence	Excessively confident	Incorrect high-confidence outputs

A learning-curve graph is useful for debugging. If training loss decreases continuously while validation loss begins to rise, the network is overfitting. Cross-validation across different regions or time periods can provide a more realistic estimate of performance.

5. Regularization Technique 1: Dropout

Dropout is a regularization technique in which a random percentage of neurons is temporarily switched off during each training step. For example, a dropout rate of 0.3 means that about 30% of selected neuron outputs are ignored during that step.

Because different neurons are removed at different times, the network cannot depend heavily on one specific pathway. It is forced to learn more distributed and stable features. During testing, all neurons are used, but their outputs are adjusted automatically.

Dropout is suitable for the rare-disease dataset because it reduces co-adaptation between neurons and prevents a small number of training records from dominating the model. A moderate dropout rate such as 0.2 to 0.5 can be tested. Very high dropout should be avoided because it may cause underfitting when the dataset is already small.

6. Regularization Technique 2: L2 Regularization

L2 regularization, also called weight decay, adds a penalty to the loss function when the network weights become too large. The modified objective can be represented as:

$$\text{Total Loss} = \text{Prediction Loss} + \lambda \sum w^2$$

Here, w represents the model weights and λ controls the strength of the penalty. The network is encouraged to use smaller and smoother weights rather than creating extreme decision boundaries.

For small outbreak datasets, L2 regularization is useful because it reduces model complexity without removing input information. It makes predictions less sensitive to noise, missing values and small changes in health indicators. The value of λ should be selected using validation data or cross-validation. A very large value may oversimplify the model, while a very small value may not control overfitting.

7. Comparison of the Two Techniques

Technique	Main Idea	Benefit	Possible Limitation
Dropout	Randomly disables neurons during training	Prevents dependency on a few neurons	High dropout may reduce learning
L2 Regularization	Penalizes large model weights	Creates smoother and more stable predictions	Strong penalty may cause underfitting

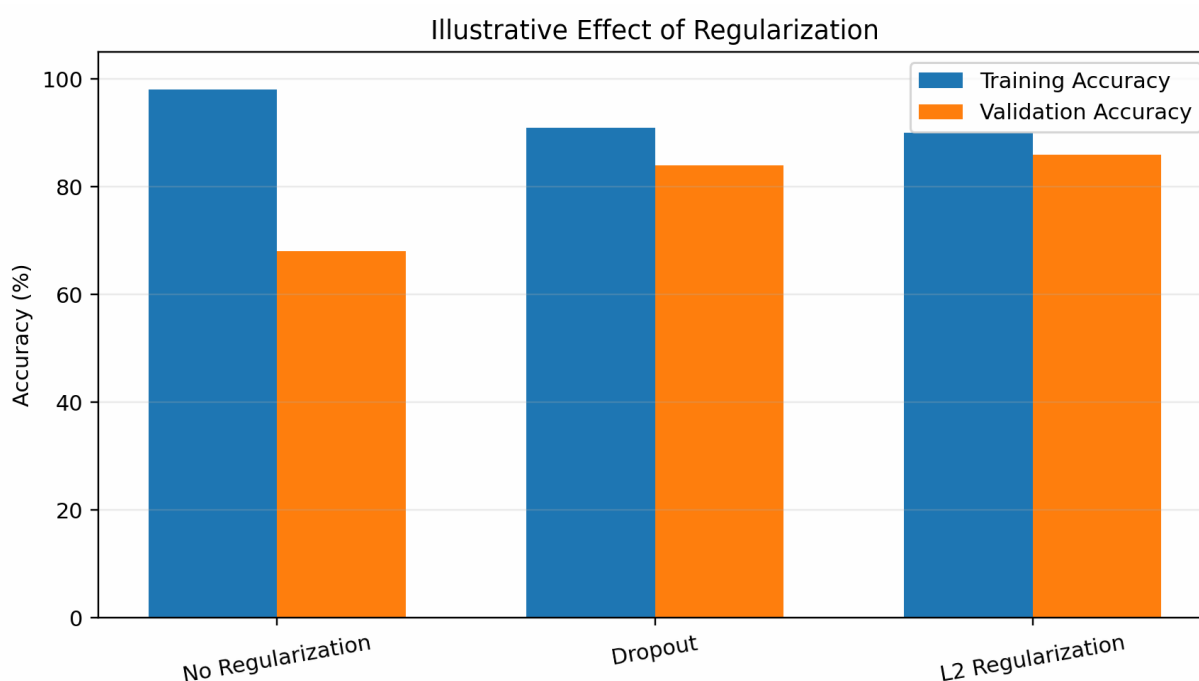


Figure 1. Illustrative comparison of training and validation accuracy

Interpretation: The unregularized model may produce very high training accuracy but much lower validation accuracy. Dropout and L2 regularization usually reduce the training score slightly while improving validation performance, which indicates better generalization. The displayed values are illustrative rather than clinical results.

8. Recommended Debugging and Training Strategy

- Split the data carefully into training, validation and testing sets. Use patient-level, region-level or time-based splitting to prevent leakage.
- Use stratified splitting so that rare outbreak cases are represented in every subset.
- Plot training and validation loss after every epoch.
- Apply early stopping and stop training when validation loss no longer improves.
- Begin with a small neural network and increase complexity only when necessary.
- Tune dropout rate and L2 penalty using cross-validation.
- Use class weights or carefully controlled resampling to handle the rare outbreak class.
- Evaluate recall, precision, F1 score and area under the precision-recall curve rather than relying only on accuracy.
- Check model calibration because health authorities need trustworthy probability estimates.
- Repeat experiments with different random seeds to confirm that the result is stable.

9. Proposed Workflow

Workflow for Reliable Rare-Disease Outbreak Prediction

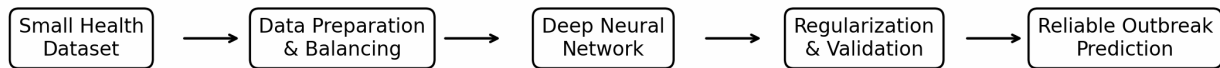


Figure 2. Workflow for reliable rare-disease outbreak prediction

The process begins with collecting a small but relevant health dataset. The data is cleaned, balanced and divided without leakage. A compact neural network is trained with dropout and L2 regularization. Validation and early stopping are then used to select the best model. Finally, the model provides an outbreak-risk score to support public-health decisions.

10. Suitability for Low-Resource Regions

The proposed approach must remain practical for regions with limited computing infrastructure, incomplete records and fewer trained health workers. A smaller neural network with dropout and L2 regularization needs less memory and computational power than a very deep model. It can be trained centrally and deployed through a lightweight web or mobile application.

- Use only essential features that are easy to collect, such as rainfall, temperature, symptom counts, water quality, vaccination coverage and population density.
- Allow missing-value handling because some reports may be unavailable.
- Provide simple Low, Medium and High risk levels instead of complex technical outputs.
- Display the most important warning factors to help local health workers understand the prediction.
- Protect patient privacy by removing names and personally identifiable information.
- Treat the model as a decision-support tool, not as a replacement for doctors or public-health authorities.

11. Evaluation Measures

Metric	Meaning	Importance for Rare Outbreaks
Recall	Percentage of actual outbreaks detected	Missing an outbreak can cause severe health impact
Precision	Percentage of warnings that are correct	Reduces unnecessary use of limited resources
F1 Score	Balance between precision and recall	Useful for imbalanced datasets
PR-AUC	Performance across alert thresholds	More informative when outbreaks are rare
Calibration	Agreement between predicted and actual risk	Supports trustworthy decision-making

12. Example Scenario

Suppose a low-resource district has only a few years of records for a rare mosquito-borne disease. A deep neural network without regularization achieves 98% training accuracy but only 67% validation accuracy. This indicates that it has memorized the limited historical cases.

After adding a dropout layer and an L2 penalty, training accuracy decreases to about 90%, while validation accuracy improves. More importantly, recall for the outbreak class increases. The regularized model is therefore more useful because it identifies future high-risk periods more reliably, even though its training accuracy is lower.

13. SDG 3 Contribution

This work supports SDG 3 - Good Health and Well-being by improving early warning for rare-disease outbreaks. Reliable predictions can help health departments prepare medicines, testing kits, hospital beds and awareness campaigns before the number of cases becomes severe.

- Supports early detection and timely response.
- Improves allocation of limited healthcare resources.
- Reduces preventable illness and death.
- Strengthens disease surveillance in underserved areas.
- Promotes responsible and equitable use of artificial intelligence in healthcare.

14. Limitations

Regularization cannot fully solve problems caused by extremely poor or biased data. If outbreak cases are incorrectly reported or some communities are not represented, the model may still produce unfair or inaccurate predictions. The dataset should therefore be reviewed by public-health experts, and the system should be tested across different regions and seasons.

Predictions must also be combined with laboratory confirmation, epidemiological investigation and local medical knowledge. A false negative may delay intervention, while too many false positives may waste scarce resources.

15. Conclusion

Overfitting in deep neural networks occurs when the model memorizes a small training dataset and fails to generalize to unseen health situations. It is likely in rare-disease outbreak prediction because the number of positive cases is limited and the available records may be noisy or imbalanced.

Dropout and L2 regularization are two suitable solutions. Dropout prevents the network from depending on particular neurons, while L2 regularization discourages excessively large weights and creates smoother decision boundaries. Together with careful data splitting, early stopping, cross-validation and appropriate evaluation metrics, these techniques can produce a more reliable and resource-efficient outbreak prediction system that supports SDG 3.

16. Suggested References

- United Nations Sustainable Development Goal 3 - Good Health and Well-being.